



312/1
GEOGRAPHY
PAPER 1
TIME: 2 3/4 HOURS

Kenya Certificate of Secondary Education (K.C.S.E.)

312/1
Geography
Paper 1
2 ¾ Hours

INSTRUCTIONS TO CANDIDATES

- This contains **two** sections **A** and **B**.
- Answer all questions in section **A**, In section **B** questions **6** and **two** other question.
- All answers must be written on booklet provided.

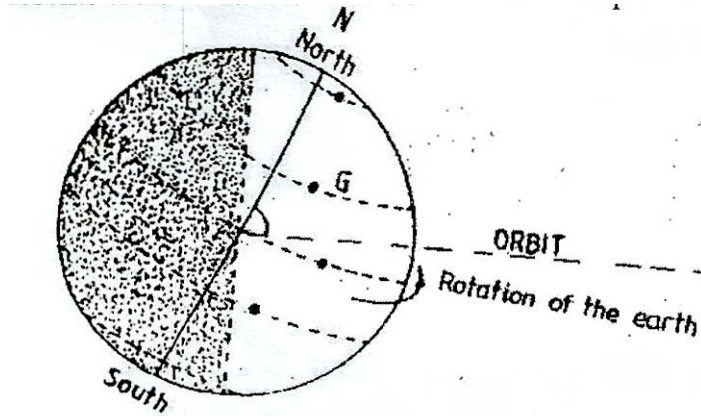
This paper consists of 4 printed pages. Candidates should check to ascertain that all pages are printed as indicated and that no questions are missing.

SECTION A

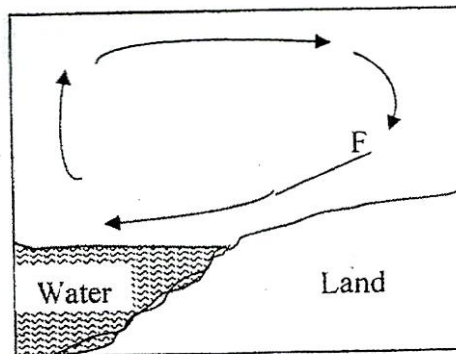
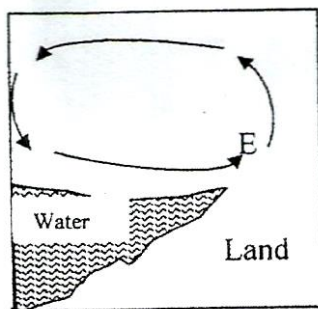
Answer all questions in this section.

1. a) Give **three** examples of mechanically formed sedimentary rocks. (3mks)
- b) State **two** changes that occur in sedimentary rocks when they are subjected to intense heat and pressure. (2mks)

2. The diagram below represents the earth on its axis. Use it to answer question (a)

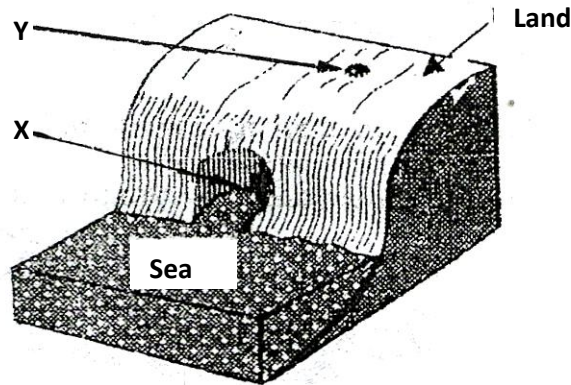


- (a) (i) Name the latitude marked **G** (1 mk)
 - (ii) What is the angle of inclination of the earth's axis from its orbit? (1 mk)
 - (b) State **three** ways that make the earth as a planet different from other planets (3mks)
3. (a) Name **three** external land forming processes that lead to the formation of lakes (3mks)
 - (b) State **three** ways in which lakes influence the natural environment (3 mks)
4. The diagram below represents the flow of air current. Use them to answer question



- (a) In your answer booklet, name the air current marked **E** and **F** (2 mks)
- (b) Give the reason why air cools as it rises (2mks)

5. The diagram below represents a coastal landform



- (a) Name the features marked **X** and **Y** (2mks)
 (b) State **three** conditions necessary for the formation of a sand spit. (3mks)

SECTIONS B

Answer questions 6 and any other two questions from this section.

6. Study the map of Karatina 1:50,000 (sheet 121/3) provided and answer the following questions.
- (a) (i) Give **two** methods used to show relief on the map (2mks)
 (ii) Give the names of the adjoining map sheets 120/4 and 135/2 (2mks)
 (iii) Convert the map scale into a statement scale. (1mk)
 (iv) State the height of the highest point on the area covered by the map. (2mks)
 (v) Calculate the area of the forest in Kirinyaga District (give your answer in square kilometers) (2mks)
- (b) (i) Name **two** human features found in grid square 8052 (1mk)
 (iii) Draw a rectangle measuring 16cm by 12 cm to represent the area enclosed by Easting's 97 to 05 and Northings 45 to 51 (1mk)
- On the rectangle mark and name the following features:
 - The tree swamp (1mk)
 - The district boundary (1mk)
 - All weather road (bound surface) to Kirinyaga (1mk)
- (c) Describe the drainage of the area covered by the map (4mks)
 (d) Citing evidence from the map, explain **three** factors that influence the distribution of settlements in the area covered by the map. (6mks)
7. (a) (i) A part from the Rift Valley name **two** other relief features that were formed as a result of faulting. (2mks)
 (ii) With the aid of a well labelled diagram, describe how a rift Valley is formed by tensional forces. (8mks)
- (b) Explain **four** effects of faulting (8mks)
 (c) Students are planning to carry out a field study of an area affected by faulting.
 (i) State **four** reasons why it is important for the students to have a pre-visit of the area (4mks)



- (ii) One of the ways they would use to collect data is through direct observation. Give **three** disadvantages of direct observation in the study of such an area. (3mks)
8. (a) (i) What is glaciations? (1mk)
 (ii) Give **three** differences between glacial trough and river valley (3mks)
- (b) (i) State **two** ice ages (4mks)
 (ii) Describe **two** ways in which ice moves (4mks)
- (c) (i) State **two** characteristics of a pyramid peak (2mks)
 (ii) Name **three** types of Maraines (3mks)
- (d) With a well labeled diagrams, describe the processes involved in the formation of a corrie lake. (6mks)
- (e) Outline **four** ways in which glaciated landscape is of negative significance to human activities. (4mks)
9. (a) Describe how a river erodes its channels by the following process.
 (i) Hydraulic action (2mks)
 (ii) Abrasion (2mks)
- (b) (i) Explain **three** factors that lead to rejuvenation of a river (6mks)
 (ii) Describe the process of a river capture. (6mks)
- (c) Some students carried out a field study on the feature found along a river.
 (i) List **three** features formed as a result of river erosion. (3mks)
 (ii) State **three** methods that students may have used to record their data (2mks)
 (iii) Explain **two** ways in which features resulting from river deposition are of significance to human activities. (4mks)
10. (a) (i) Differentiate between weathering and denudation (2mks)
 (ii) State **three** factors that influence the rate at which weathering occurs. (3mks)
- (b) Explain how the following processes of weathering occur in arid areas.
 • Crystal growth (2mks)
 • Exfoliation (2mks)
 • Block disintegration (2mks)
 • Granular disintegration (2mks)
- (c) (i) Apart from soil creep, name **three** other types of slow-mass wasting (3mks)
 (ii) State **three** causes of soil creep (3mks)
- (d) Students from a school in Rachuonyo South sub county carried out a field study on land slides.
 (i) Identify **two** ways they must have prepared for the field study. (2mks)
 (ii) Name **two** types of rapid mass movement they could have identified. (2mks)
 (iii) State **two** effects of rapid-mass wasting they came across. (2mks)